

MEMO+ project

Monitoring of exotic mosquitoes in Belgium
Results of the 2025 surveillance



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Key points

- In 2025, 599 notifications of potential tiger mosquitoes were received via the citizen science platform MosquitoSurveillance. Among these, 48 notifications were actual tiger mosquitoes.
- The notifications of tiger mosquitoes originated from nine different municipalities, most of them in Flanders (7), but also one in Wallonia and one in Brussels-capital region. Of the nine municipalities, four were new, meaning that no notifications had been recorded in previous years.
- An additional notification of tiger mosquito was made via the platform observations.be in a new municipality in Brussels-capital region.
- In seven out of the nine municipalities where mosquitoes were reported via MosquitoSurveillance, field inspections were carried out. Pupae, larvae and adult mosquitoes were found during the inspection in two of these seven municipalities.
- Potential overwintering was confirmed in three locations (Hoegaarden, Sint-Joost-ten-Node and Wijnegem).
- In Hoegaarden the longitudinal monitoring revealed persistence and spread of the species.
- Other exotic mosquitoes, *Aedes japonicus* (30 reports) and *Aedes koreicus* (6 reports) were also reported through the citizen science platform.
- Since the start of the MEMO+ project we received at least one tiger mosquito notification from 39 municipalities (40 including the notification in Watermael-Boitsfort via observations.be) through the citizen science platform MosquitoSurveillance, with introductions detected in new municipalities every year.
- Over time, we observed that the number of municipalities where tiger mosquitoes were observed in several years is increasing and that in some known municipalities, notifications are received from several locations.
- The large number of notifications from citizens shows, once again, that citizen surveillance is an important asset in the detection of new locations and that in some cases it supports the follow-up of the situation in known locations.
- The overall results of these four years monitoring clearly shows that the tiger mosquito keeps entering and expanding its territory in Belgium.
- The involvement of the public and local authorities in surveillance efforts remains essential to properly monitor the situation in Belgium.
- The results of the surveillance support further possible actions for prevention and control of the tiger mosquito.

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1. Introduction

The introduction of exotic *Aedes* mosquitoes such as the tiger mosquito (*Aedes albopictus*) poses a potential threat to public health in Europe and Belgium. Aside of causing strong nuisance due to his diurnal biting behavior, the tiger mosquito can be vector of several viruses such as chikungunya, dengue and zika. The tiger mosquito is established in many countries around the Mediterranean and over the past two decades, the species has expanded its range northward in Europe. It is now established in almost all departments in France, in some regions of Germany as well as in some Belgian municipalities. In some of the countries where it is established since years, the tiger mosquito has been responsible for local transmission of chikungunya and dengue, with outbreaks reported every year. To date, no local transmission chikungunya, dengue or zika have been reported in Belgium.

In order to be prepared and to minimize the risk of potential local transmission of *Aedes*-borne viruses in the future, it is essential to monitor the presence of the tiger mosquito in Belgium. In this context, the MEMO+ project started in 2022. The project is responsible for monitoring exotic mosquitoes in Belgium, with a specific focus on the tiger mosquito, combining citizen surveillance and field inspections:

- **Citizen surveillance:** citizens can report the presence of *Ae. albopictus* through pictures via the citizen science website www.mosquitosurveillance.be or the mobile app ([IOS](#) ou [android](#)).
- **Field inspections:** which consisted in 2025 of three activities:
 - the field inspection after a notification of *Ae. albopictus* by a citizen;
 - the surveillance of five potential overwintering sites;
 - the longitudinal surveillance of presence and spread at two locations.

The MEMO+ project is coordinated by Sciensano, in collaboration with the Institute of Tropical Medicine, Antwerp. Besides the overall coordination, Sciensano is also responsible for the coordination of the citizen science, while the Institute of Tropical Medicine coordinates and carries out the field inspections. This report presents the results of the fourth year of the MEMO+ project, covering the period from January to December 2025.

2. Results of the surveillance

2.1. Overview of the citizen surveillance

In 2025, 599 notifications of potential tiger mosquitoes in Belgium were made by citizens via the MosquitoSurveillance platform. In addition, 31 notifications were coming from other countries i.e., France, Italy, Morocco, Slovenia, Spain and the Netherlands. Of the total of 599 notifications from Belgium, 358 (59.8%) were made in Flanders, 184 (30.7%) in Wallonia, and 38 (6.3%) notifications were made in Brussels. Another 19 (3.2%) were lacking a postal code. The majority (501) of the notifications were made between May and September. One major peak of notifications was observed during the week with the press release in August (**Figure 1**). Notifications were received from the ten provinces of Belgium.

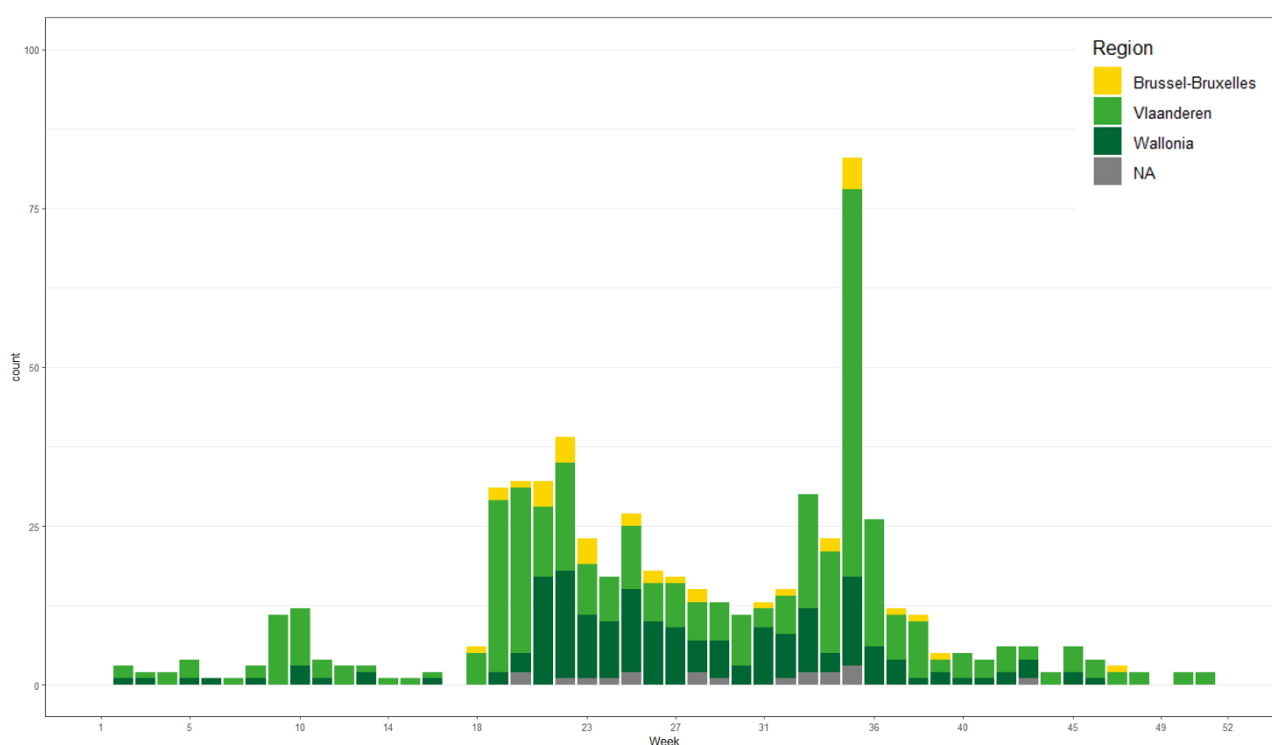


Figure 1 Number of notifications of potential tiger mosquitoes via MosquitoSurveillance per week and per region in 2025.

Of the total notifications (599), 542 (90.5%) were of an insect, and of these, 382 (70.5%) were identified as a mosquito (Culicidae). Among the Culicidae, we identified the following genera: *Aedes* spp., *Anopheles* spp., *Culiseta* spp. and *Culex* spp. (**Tableau 1**).

Tableau 1 Overview of the notifications made through MosquitoSurveillance in 2025.

	Number (%)
Total number of notifications	599
Pictures of an insect	542 (90.5%)
Insects	Number (% of the insects)
Culicidae (= mosquito)	382 (70.5%)
Other insects	151 (27,9%)
Unidentifiable (insect or mosquito)	9 (1,7%)
Culicidae	Number (% of the Culicidae)
<i>Aedes</i>	148 (38.8%)
<i>Anopheles</i>	2 (0.5%)
<i>Culex</i>	7 (1.8%)
<i>Culiseta</i>	197 (51.6%)
No genus defined (Culicidae)	28 (7.3%)

2.2. Notifications of tiger mosquitoes and inspections

2.2.1. Notifications of tiger mosquitoes

In 2025, we received 48 notifications of tiger mosquitoes (*Ae. albopictus*) i.e., 12.6% of the 382 Culicidae notifications. The first notification of *Ae. albopictus* was made on 11 May (week 19), the second one on 4 June (week 23), and the third on 18 July (week 29). From week 29 until week 40, at least one notification of *Ae. albopictus* was made every week (except for week 32 and week 39). The last notification of the season was made in week 44 (**Figure 2**). The 48 notifications of *Ae. albopictus* came from nine different municipalities spread over the three regions in Belgium. Of these nine municipalities, four were new, meaning that a tiger mosquito was never observed there in the past. The distribution of tiger mosquito notifications per province and municipalities can be found in **Tableau 2**.

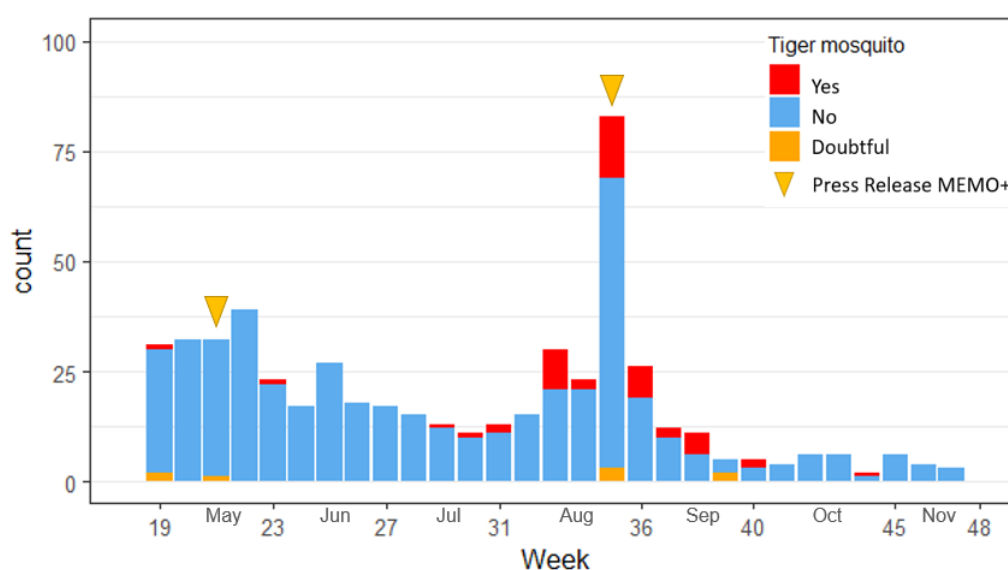


Figure 2 Number of tiger mosquito notifications via MosquitoSurveillance per week from May to November 2025 (19-48).

Tableau 2 Overview of the number of tiger mosquitoes reported by citizens in 2025 per province and municipality.

PROVINCE/ Municipality	Number of tiger mosquitoes	New municipality
ANTWERP	3	
Wijnegem	3	No
EAST FLANDERS	19	
Heusden (Destelbergen)	1	Yes
Merelbeke-Melle	18	No
FLEMISH BRABANT	22	
Kortenberg	1	Yes
Zaventem	9	Yes
Kessel-Lo	2	No
Hoegaarden	10	No
BRUSSELS Capital Region	1	
Etterbeek	1	Yes
HAINAUT	3	
Ath	3	No

In addition, *Ae. albopictus* was also observed in another new municipality, Watermael-Boitsfort. This notification was made through 'Waarnemingen.be', a biodiversity platform for citizen science and was therefore not included in the numbers and figures of this report.

2.2.2. Field inspection after a notification of tiger mosquito

In total, eight field inspections in seven municipalities were conducted in 2025 following reports of citizens. In four municipalities, namely Kessel-Lo, Kortenberg, Merelbeke-Melle, Zaventem (two inspections) the pictures sent by the citizens clearly showed a tiger mosquito. For the three other municipalities where a field inspection was carried out – namely Beyne-Heusay, Maaseik and Maasmechelen – it was difficult to say with certainty, based only on the pictures accompanying the report, that the mosquito in question was a tiger mosquito. The mosquito from the report in Maasmechelen had been preserved and could later be identified with certainty as *Aedes koreicus* via DNA-barcoding. For the reports from Beyne-Heusay and Maaseik, the reported mosquitoes had not been retained, so we were unable to carry out further identification. Both reports were classified as 'doubtful' for *Ae. albopictus*.

The inspections were carried out during one day at the notifier's garden, along with a selection of randomly chosen gardens within a 100-meter radius around the first location of a notification. Two field technicians were looking for larvae in potential breeding sites in gardens. If adult mosquitoes were observed during the inspection, they would be collected by hand. In two (Merelbeke-Melle and one inspection of Zaventem) out of the eight inspections, the field work confirmed the presence of *Ae. albopictus* as larvae, adults, or both (see **Tableau 3** for more details).

2.2.3. Overwintering monitoring

During the 2024 season, field inspections and notifications received from citizens revealed a high number of tiger mosquitoes in Boom, Forest, Sint-Joost-ten-Node, Wijnegem and Schelle. In order to investigate whether this population survived the winter, researchers executed field inspections at the beginning of May and June 2025 to inspect for larvae and adult tiger mosquitoes that emerged from eggs. Several houses, where tiger mosquitoes were observed in 2024, were revisited twice with

a four weeks interval to perform larval samplings and to set up a trap for adult mosquito collection. At two locations (Wijnegem and Sint-Joost-ten-Node) *Ae. albopictus* adults were collected in the trap. In addition, in Sint-Joost-ten-Node, tiger mosquito larvae were found in one garden (see **Tableau 3** for more details).

2.2.4. Longitudinal monitoring

In order to assess the start, end and peak of the active season of *Ae. albopictus*, monitoring was carried out from the end of April to November in two contrasting sites: an urban/peri-urban site (Wilrijk) versus a rural site (Hoegaarden). Twenty oviposition traps to collect eggs and two BG-sentinel traps to collect adult mosquitoes were placed in a 500m radius buffer zone. The traps were inspected every two weeks (in total 14 sampling dates in 6 months). Additionally, three inspections for larvae were executed in Hoegaarden and two in Wilrijk at the edge of the 500m buffer zone. The results are the following (see **Tableau 3** for more details).

In Hoegaarden:

- **Oviposition traps:** 515 eggs were found spread over 10 oviposition traps (out of the 20) at seven different sampling dates between half of June and end of September.
- **Larval inspections:** immature stages (larvae and pupae) were detected during one larval inspection at the end of Aug. No larvae or pupae were collected during the other inspections. Several adults were hand captured during two out of three larval inspections.
- **Adult traps:** adults were captured with one trap in the center of the zone at 10 different sampling dates, and in a second trap at two different sampling dates. Overall, adults were captured between the beginning of May and the end of September.

In Wilrijk:

- **Oviposition traps:** no eggs were detected
- **Larvae:** during two larval inspections, no larvae or pupae were detected
- **Adult traps:** no adults were captured.

Tableau 3 Overview of the 2025 results of the field activities. The numbers indicate; the number of places where larvae (larval sampling) and adults (hand capture) were found during the inspection, the number of oviposition traps or BG-sentinel traps with *Ae. albopictus*, the number of tiger mosquito notifications received in 2025.

Type of field inspection	Location	Field activities				Citizen surveillance
		Larval sampling*	Hand capture ^Δ	Oviposition trap ⁰	BG-Sentinel trap ^x	Number of notifications of a tiger mosquito
Inspection after a notification	Beyne-Heusay**	0	N/A	N/A	N/A	0
	Kessel-Lo	0	N/A	N/A	N/A	2
	Kortenbergh	0	N/A	N/A	N/A	1
	Maaseik**	0	N/A	N/A	N/A	0
	Maasmechelen**	0	N/A	N/A	N/A	0
	Merelbeke-Melle	6	5	N/A	N/A	18
	Zaventem n°1	4	1	N/A	N/A	9
	Zaventem n°2	0	N/A	N/A	N/A	
Overwintering	Boom	0	N/A	N/A	0	0
	Forest	0	N/A	N/A	0	0
	Sint-Joost-ten-Node	1	N/A	N/A	1	0
	Schelle	0	N/A	N/A	0	0
	Wijnegem	0	N/A	N/A	1	3
Longitudinal follow-up	Hoegaarden	1	4	9	2	10
	Wilrijk	0	N/A	0	0	0

***Larval sampling**: number of places, in most cases gardens of houses, where larvae were found; ^Δ**Hand capture**: number of places, in most cases garden of houses, where adults were found; ⁰**Oviposition trap**: number of oviposition traps in which eggs were found at least once during the longitudinal follow-up; ^x**BG-sentinel trap**: number of traps in which adults were found at least once. ** Other exotic *Aedes* mosquitoes were found during the inspection. *Ae. japonicus* in Beyne-Heusay and *Ae. koreicus* in Maaseik and Maasmechelen.

2.3. Other exotic mosquitoes

Other exotic *Aedes* mosquitoes, such as *Aedes japonicus* and *Ae. koreicus*, were reported besides *Ae. albopictus* (tiger mosquito). Unlike notifications of *Ae. albopictus*, and since the surveillance isn't currently focused on the other exotic mosquitoes, no inspection was executed after receiving a notification of those species. Of the 148 *Aedes* notifications that we received during the 2025 season, 40 were of an exotic *Aedes* species aside of *Ae. albopictus*, namely *Ae. japonicus* (n=30), *Ae. koreicus* (n=6) and four mosquitoes for which species identification wasn't conclusive between *Ae. japonicus* and *Ae. koreicus*. The 40 notifications were made between April (week 16) and October (week 43) and originated from 26 unique municipalities (NIS5 level) (**Tableau 4**). Most findings are in the eastern part of Belgium spread across the provinces of Liège (20), Limburg (10) and Luxembourg (9). In addition, there was one *Ae. japonicus* notified in the province of Antwerp.

Tableau 4 Overview of the municipalities (postcode level) where notifications of *Ae. japonicus*, *Ae. koreicus*, doubtful *Ae. japonicus/Ae. koreicus* were made via MosquitoSurveillance in 2025.

Province	Liège	Limburg	Luxembourg	Antwerp
<i>Ae. japonicus</i>	Angleur, Argenteau, Aywaille, Bassenge, Beyne-Heusay, Blégny, Burdinne, Cheratte, Grâce-Hollogne, Herstal, La Calamine, Lanaye, Trooz, Wandre	Boorseme, Hasselt, Lanaken, Maaseik	Aubange, Chiny, Fauvillers, Libin, Messancy, Musson, Neufchâteau, Robelmont, Tintigny	Balen
<i>Ae. koreicus</i>	Bassenge, Wandre	Maasmechelen, Niel-bij-As		
<i>Ae. japonicus/Ae. koreicus</i>	Blégny, Liège	Maaseik, Massmechelen		

3. Discussion & Conclusion

Since the start of the MEMO+ surveillance in 2022, we observed an increase in citizen participation from different municipalities. In 2022, we received notifications of potential tiger mosquito from 147 municipalities, in 2023 from 271 municipalities, in 2024 from 356 municipalities and in 2025 from 259 municipalities. Taking all these notifications into account, there has been, since 2022, at least one notification of a potential tiger mosquito from 474 out of the 565 municipalities (NIS5 level) in Belgium. Even though the number of notifications in 2025 was lower ($n= 599$) compared to 2024¹ ($n= 1289$), we did observe participation from an additional 22 unique municipalities from which we didn't received any notification before.

Regarding notifications of actual tiger mosquitoes, in 2025, the number was comparable to that of 2024 (respectively 48 and 47). However, these reports originated from fewer municipalities (nine municipalities in 2025 versus 21 in 2024), four of which were new. This concentration of notifications in a limited number of municipalities could be explained by the fact that of some areas facing not just isolated introductions, but local establishment and more widespread populations. This was observed for at least the second consecutive year in Merelbeke-Melle and Hoegaarden, and for the first time in Zaventem. In all three municipalities, field inspections confirmed the presence of the species in significant numbers

Since 2022, the tiger mosquito has been detected in 40 municipalities (incl. Watermael-Boitsfort) (**Figure 3**). Flanders is currently much more affected in comparison to the other regions, with 30 municipalities where *Ae. albopictus* has been reported. In Wallonia four municipalities are affected, and in Brussels, new locations are being discovered since the past two years, with a total of six municipalities where *Ae. albopictus* has been reported (incl. Watermael-Boitsfort). Over time, we also observed that in known municipalities, notifications are received from different places within a municipality (for example in Schelle, Merelbeke-Melle, Zaventem, Hoegaarden, Kessel-Lo) suggesting the spread of the species. Also, the number of municipalities where tiger mosquitoes were observed over several years is increasing.

¹ Delbecq J, Hermy MRG, Deblauwe I, Schneider A, Lernout T, Müller R, Van Bortel W, Rebolledo J. Monitoring of Exotic Mosquitoes in Belgium (MEMO+): Results of the 2024 surveillance. Brussels, Belgium. Sciensano; 2025.

This year, overwintering has been confirmed in three municipalities: Sint-Joost-ten-Node and Wijnegem, where the tiger mosquito was first reported in 2024, and in Hoegaarden, where the first sighting was reported in 2022. Overwintering has previously been confirmed in Wilrijk, Lebbeke, Ath, Puurs-Sint-Amands and Kessel-Lo bringing the total number of known locally established tiger mosquito locations in Belgium to eight. In addition, the longitudinal monitoring in Hoegaarden revealed that the tiger mosquito population has spread further beyond a distance of more than 500 meters. In Wilrijk, no tiger mosquitoes were detected throughout the season. However, in another project that also took place in Wilrijk (MASSTRAP) and where a high number of traps was placed in the center of the infestation, several adult mosquitoes were collected in the summer of 2025.

We can see that, year after year, the citizen notifications are key in the detection of new locations and that they are also an important tool to follow-up the situation in known locations. We therefore call on citizens to continue reporting any sightings of potential tiger mosquitoes via the platform [MosquitoSurveillance](#), even if they have already reported them in the past. In parallel, the execution of field work remains crucial to get a better understanding of the exact situation in each location. This information supports further possible actions for prevention and control of the tiger mosquito.

In conclusion, based on all the above information we can state that the entomological situation of the tiger mosquito in Belgium is continuously evolving with new introductions and additional established populations and spread being identified each year. Although no autochthonous arbovirus transmission has been reported to date, continued introductions of *Ae. albopictus* and its increasing establishment raise the risk of future outbreaks. Monitoring efforts should continue in order to detect the presence of the tiger mosquito and to better understand its ecology. The information that is gathered in the MEMO+ project is used to advice the competent authorities to take actions.

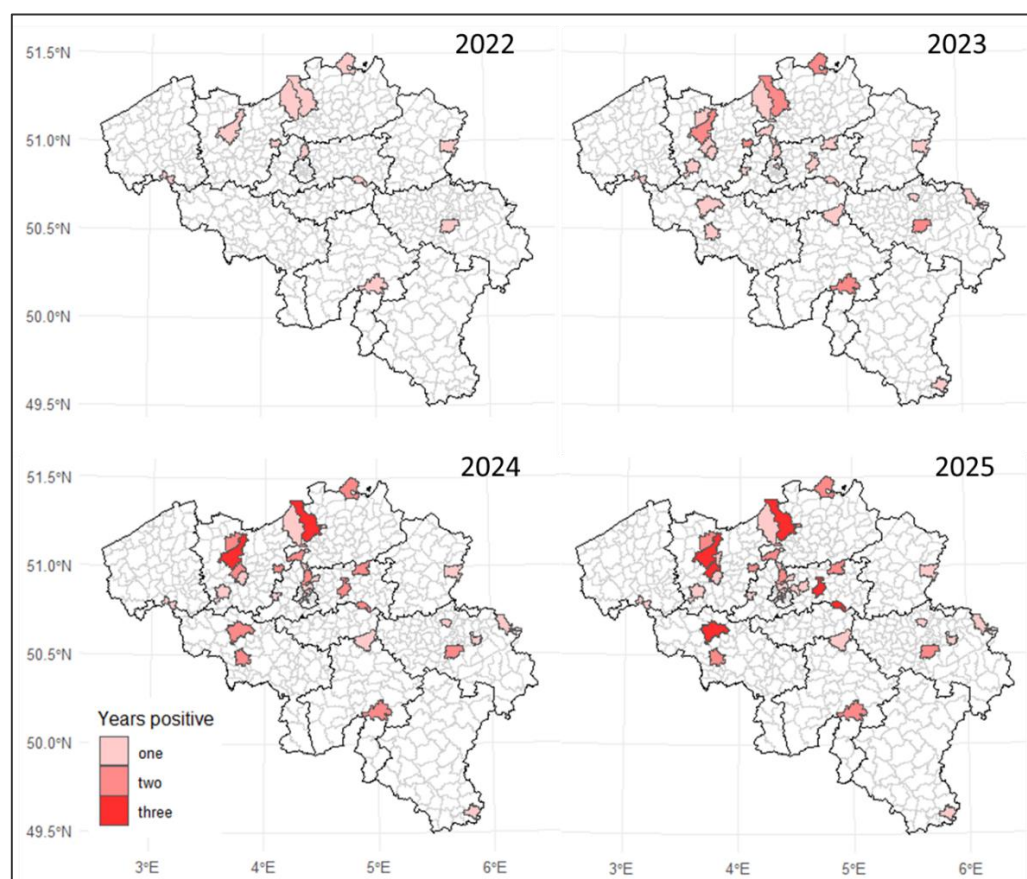


Figure 3 Representation of the number of years tiger mosquitoes have been observed in a municipality, between 2022 and 2025. These figures show results from the field inspections and citizen surveillance.

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Sciensano is a public and independent scientific institute dedicated to human and animal health and their environment.

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